Depth First Search (DFS)

Aim

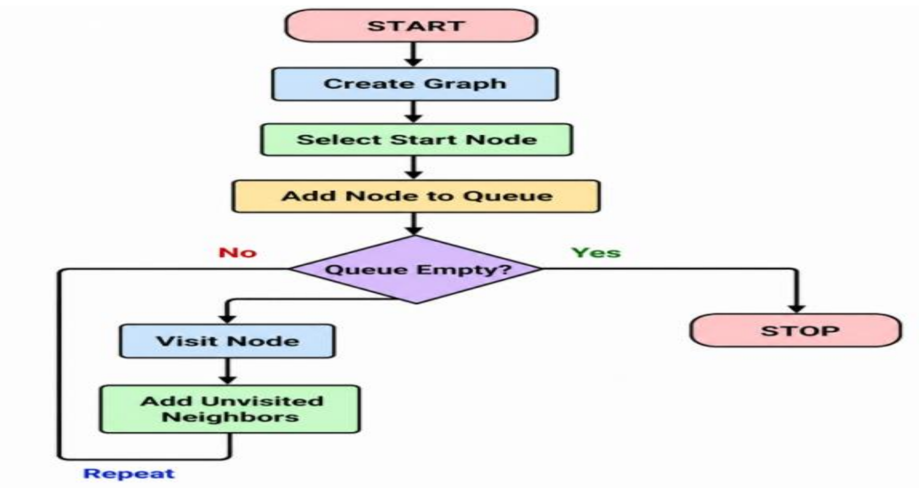
To implement the Depth First Search (DFS) algorithm for graph traversal.

Objectives

- To understand depth-wise traversal.
- To visit all graph nodes.
- To implement DFS using recursion.
- To explore each branch completely before backtracking.

Algorithm

1. Start.
2. Create a graph.
3. Select the starting node.
4. Mark the node as visited.
5. Visit the node.
6. Recursively visit all unvisited adjacent nodes.
7. Repeat until all nodes are visited.
8. Stop.



PROGRAM

```python
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbor in graph[node]:
            dfs(neighbor)

dfs('A')
```

OUTPUT

```
A B D E C F
```

Result

The DFS algorithm was successfully implemented and traversed the graph using depth-first traversal.